Monte Carlo Pricing of a European Call in a Hybrid Equity–Short Rate Model

Niccolò Maria Angelini, Massimiliano Biggio, Rosita Cirillo, Leonardo Murari, Volodymyr Pavliv, Michele Rinaldi

January 2026 - Group 11
Computational Methods in Finance - 21040

1 Introduction and problem statement

This report addresses the Monte Carlo pricing of a European call option in a hybrid model combining a lognormal equity dynamics with a stochastic short interest rate. Under the risk–neutral probability measure $\mathbb{Q}$, the equity price S_t and the short rate r_t evolve according to

$$\begin{cases} \dfrac{dS_t}{S_t} = r_t\,dt + \sigma_S\,dW_S^{\mathbb{Q}}(t), \\ dr_t = \kappa(\theta - r_t)\,dt + \sigma_r\,dW_r^{\mathbb{Q}}(t), \end{cases} \tag{1}$$

where $\sigma_S > 0$ and $\sigma_r > 0$ are volatilities, $\kappa > 0$ is the mean-reversion speed, θ is the long-run mean level of the short rate, and $(W_S^{\mathbb{Q}}, W_r^{\mathbb{Q}})$ are correlated Brownian motions with instantaneous correlation coefficient $\rho \in [-1, 1]$, i.e.

$$d\langle W_S^{\mathbb{Q}}, W_r^{\mathbb{Q}}\rangle_t = \rho\,dt.$$

The contract is a European call option with maturity T and strike K, written on the equity S. Its arbitrage-free time–0 premium under $\mathbb{Q}$ is given by the risk-neutral discounted expectation

$$C_0 = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^T r_t\,dt\right)(S_T - K)^+\right], \tag{2}$$

where $(x)^+ = \max(x, 0)$. The objectives are (i) to compute C_0 by Monte Carlo simulation using a discretization scheme consistent with the model structure, and (ii) to validate the estimate by comparison with a closed-form pricing formula available for this hybrid specification.

2 Closed-form benchmark

For the model (1), the call premium admits an analytical expression of Black–Scholes type:

$$C_0 = S_0\,N(\tilde{d}_1) - K\,p(0,T)\,N(\tilde{d}_2), \tag{3}$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function, $p(0,T)$ is the time–0 price of a zero-coupon bond maturing at T, and

$$\tilde{d}_1 = \frac{\ln\left(\frac{S_0}{Kp(0,T)}\right) + \frac{1}{2}\Sigma^2}{\sqrt{\Sigma^2}}, \qquad \tilde{d}_2 = \tilde{d}_1 - \sqrt{\Sigma^2}. \tag{4}$$

2.1 Zero-coupon bond price under Vasicek

Under the Vasicek short-rate dynamics in (1), the zero-coupon bond price can be written in exponential-affine form

$$p(0,T) = A(T)\, e^{-B(T) r_0}, \tag{5}$$

where

$$B(T) = \frac{1-e^{-\kappa T}}{\kappa}, \qquad A(T) = \exp\left[\left(\theta - \frac{\sigma_r^2}{2\kappa^2}\right)(B(T)-T) - \frac{\sigma_r^2}{4\kappa} B(T)^2\right]. \tag{6}$$

2.2 Effective variance

The quantity Σ^2 in (4) is the effective variance that incorporates equity volatility, stochastic discounting effects and the instantaneous correlation ρ between the equity and rate shocks:

$$\Sigma^2 = \sigma_S^2 T + \frac{2\sigma_r \sigma_S \rho}{\kappa^2}\left(\kappa T - 1 + e^{-\kappa T}\right) - \frac{\sigma_r^2}{2\kappa^3}\left(3 + e^{-2\kappa T} - 4e^{-\kappa T} - 2\kappa T\right). \tag{7}$$

Equation (3) is used as a benchmark to assess the accuracy of the numerical method.

3 Monte Carlo methodology

3.1 Time grid and correlated innovations

The interval $[0,T]$ is discretized into m equal steps with $\Delta t = T/m$ and grid times $t_i = i\Delta t$ for $i = 0,\dots,m$. At each step, two independent standard normal random variables $Z_1, Z_2 \sim \mathcal{N}(0,1)$ are generated and mapped into correlated Brownian increments through

$$\Delta W_S = \sqrt{\Delta t}\, Z_1, \qquad \Delta W_r = \sqrt{\Delta t}\left(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\right). \tag{8}$$

This construction guarantees $\mathrm{Corr}(\Delta W_S, \Delta W_r) = \rho$ at each time step.

3.2 Discretization of the short rate

The short rate is simulated by an Euler scheme:

$$\hat{r}_{t_i} = \hat{r}_{t_{i-1}} + \kappa(\theta - \hat{r}_{t_{i-1}})\Delta t + \sigma_r\, \Delta W_r, \qquad i = 1,\dots,m. \tag{9}$$

A relevant feature of the Vasicek specification is that it is Gaussian and therefore does not impose positivity of interest rates; negative simulated values can occur and are consistent with the model assumptions.

3.3 Conditionally exact step for the equity

Conditionally on the short rate being approximated as constant on $[t_{i-1}, t_i]$ at the value $\hat{r}_{t_{i-1}}$, the equity update admits the exact lognormal form:

$$\hat{S}_{t_i} = \hat{S}_{t_{i-1}} \exp\left[\left(\hat{r}_{t_{i-1}} - \frac{1}{2}\sigma_S^2\right)\Delta t + \sigma_S\, \Delta W_S\right], \qquad i = 1,\dots,m. \tag{10}$$

This step reduces discretization error relative to a direct Euler update for S_t and is natural in a lognormal model where the drift depends on the short rate.

3.4 Pathwise discounting and Monte Carlo estimator

For each simulated path $j = 1, \ldots, N$, the discounted payoff is computed as

$$\hat{V}^{(j)} = \hat{D}^{(j)}(\hat{S}_T^{(j)} - K)^+, \qquad \hat{D}^{(j)} = \exp\left(-\sum_{i=0}^{m-1} \hat{r}_{t_i}^{(j)} \Delta t\right),$$

where the integral $\int_0^T r_t \, dt$ is approximated by a left Riemann sum on the simulated grid. The Monte Carlo estimator of the option price is then

$$\hat{C}_0 = \frac{1}{N} \sum_{j=1}^{N} \hat{V}^{(j)}. \tag{11}$$

The sampling uncertainty is quantified by the standard error

$$\widehat{\mathrm{SE}} = \frac{\mathrm{Std}(\hat{V}^{(1)}, \ldots, \hat{V}^{(N)})}{\sqrt{N}},$$

yielding an approximate 95% confidence interval $\hat{C}_0 \pm 1.96\, \widehat{\mathrm{SE}}$.

4 Financial, mathematical and numerical issues

The principal numerical challenge is the coexistence of two error sources: discretization error and Monte Carlo sampling error. The rate process is discretized with Euler, which introduces a time-step bias that decreases as $\Delta t \to 0$. This bias affects the option price through both the simulated drift of the equity (via r_t in (1)) and, more importantly, through the stochastic discount factor in (2). The equity process is instead handled through the lognormal update (10), which is exact conditional on the left-endpoint approximation of r_t over each time step. This significantly reduces discretization error in S_t and leaves the rate discretization as the dominant time-grid effect.

From a financial perspective, even though the payoff depends only on S_T, the premium depends on the joint law of $(S_T, \int_0^T r_t \, dt)$ under $\mathbb{Q}$. Correlation plays a non-trivial role because the equity shock and the rate shock interact in two ways: by affecting the distribution of S_T through the drift term and by affecting discounting through the covariance between S_T and the random discount factor. This explains why the option value is sensitive to ρ even for a plain-vanilla payoff.

A further modelling aspect is the possibility of negative rates in the Vasicek model. While potentially unrealistic in some environments, it is a known feature of Gaussian short-rate models and is acceptable over moderate horizons, particularly when the purpose is to study hybrid pricing mechanisms and validate a numerical scheme against an analytical benchmark.

5 Implementation and programming considerations

The simulation is implemented in a vectorized form to leverage array operations and reduce computational time. For fixed m and N, two arrays of size $(N, m+1)$ store the simulated paths of r and S. This choice allows efficient computation of the pathwise integral $\sum_{i=0}^{m-1} \hat{r}_{t_i} \Delta t$ and the payoff evaluation at maturity. The primary trade-off is memory usage; however, for the selected parameters ($N = 100{,}000$ and $m = 250$), the memory footprint remains manageable in a cloud notebook environment.

Reproducibility is achieved by fixing the random seed. In the sensitivity analysis, resetting the seed inside each pricing function produces comparable estimates across different parameters by using common random numbers; this reduces the variability of differences and makes trends with respect to σ_r and ρ easier to interpret.

Correct handling of correlation is essential. The construction (8) ensures the intended instantaneous correlation at the discretization level. Particular care is required when exploring extreme values of ρ near ± 1, where numerical round-off could make $1-\rho^2$ slightly negative; this is avoided in the reported experiments by using values within the admissible range and standard floating-point precision.

6 Numerical results and validation

The baseline simulation uses a one-year horizon and daily discretization ($T = 1$, $m = 250$) with $N = 100{,}000$ Monte Carlo paths. The parameters are set to

$$r_0 = 0.0364, \quad \theta = 0.04, \quad \kappa = 0.2, \quad \sigma_r = 0.02, \quad S_0 = 140, \quad \sigma_S = 0.25, \quad K = 100, \quad \rho = 0.25.$$

The Monte Carlo estimator yields

$$\hat{C}_0 = 44.475234, \qquad 95\% \text{ CI } \approx [44.2622,\ 44.6883].$$

The closed-form benchmark (3)–(7) gives

$$C_0^{\text{closed}} = 44.501294.$$

The analytical value lies within the Monte Carlo confidence interval, indicating that the numerical scheme is correctly implemented and that discretization bias at $\Delta t = 1/250$ is small relative to sampling uncertainty.

6.1 Discussion of the (small) discrepancy between Monte Carlo and closed form

The closed-form premium (3) is obtained in continuous time under the exact model dynamics (1), whereas the Monte Carlo estimator necessarily relies on time discretization. In particular, the simulation produces a discretized pair $(\hat{S}_{t_i}, \hat{r}_{t_i})$ and replaces the continuous-time price (2) with the discrete-time functional

$$C_0^{\Delta t} = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\sum_{i=0}^{m-1} \hat{r}_{t_i}\Delta t\right)\left(\hat{S}_T - K\right)^+\right], \tag{12}$$

where $\int_0^T r_t\,dt$ is approximated by a left Riemann sum and r_t is simulated via the Euler scheme (9). As $\Delta t \to 0$ the approximation becomes exact (in the weak sense relevant for pricing), but for finite Δt the estimator exhibits a small numerical bias. In practice, this bias is mainly driven by: (i) the Euler discretization of the short-rate process, which slightly alters the finite-step distribution of r_t and therefore the joint law of $(S_T, \int_0^T r_t dt)$; and (ii) the discretization of the stochastic discount factor, which uses $\exp(-\sum \hat{r}_{t_i}\Delta t)$ instead of $\exp(-\int_0^T r_t dt)$.

A related point concerns the nonlinearity of discounting. Let $X = \sum_{i=0}^{m-1} \hat{r}_{t_i}\Delta t$ denote the (discretized) integrated short rate. Since the exponential function is convex, Jensen's inequality implies

$$\mathbb{E}\left[e^{-X}\right] \ \geq\ e^{-\mathbb{E}[X]}. \tag{13}$$

Therefore, replacing stochastic discounting by discounting at the average rate (i.e., using $e^{-\mathbb{E}[X]}$ in place of $\mathbb{E}[e^{-X}]$) would systematically *underestimate* the average discount factor and, consequently, the option value. In our implementation we correctly discount path-by-path via e^{-X} as in (12); the remaining difference observed between Monte Carlo and closed form is thus attributed to discretization effects (which vanish as $\Delta t \to 0$) together with the finite-sample Monte Carlo error quantified by the confidence interval.

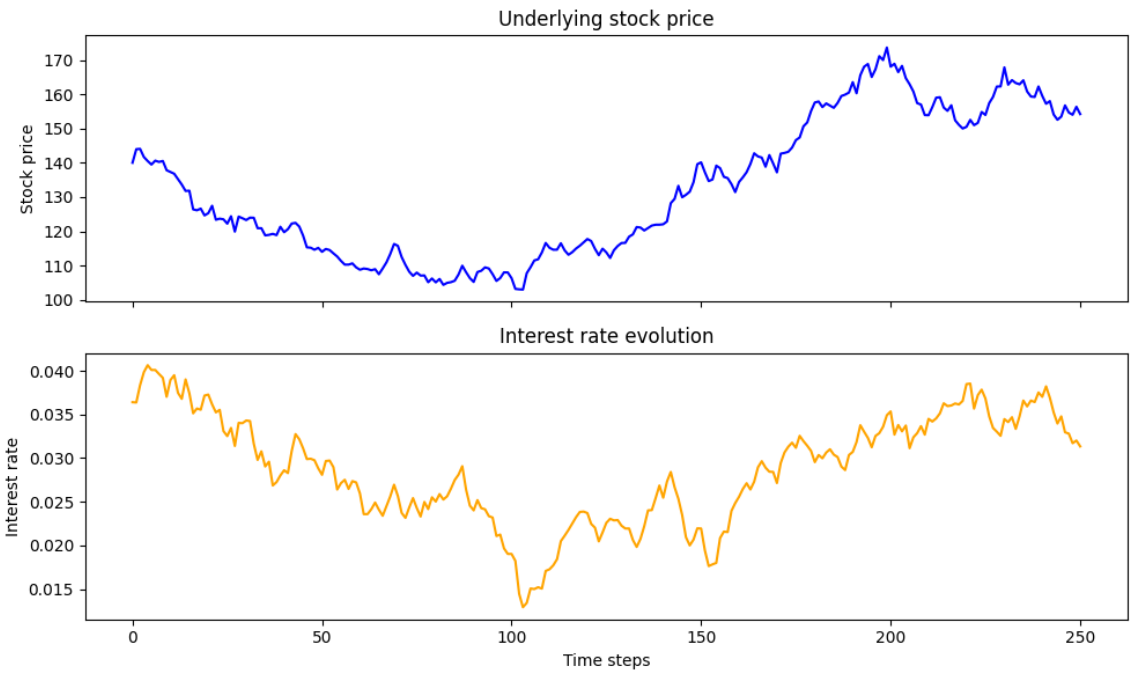

Figure 1: One simulated path for the equity price and the short rate under the hybrid model. The short rate is simulated via Euler discretization, while the equity follows a lognormal step conditional on the simulated rate.

7 Sensitivity analysis

The sensitivity study focuses on the impact of the interest-rate volatility σ_r and the correlation ρ on the call premium. Keeping the other parameters fixed, the price is computed across grids of σ_r and ρ using the same Monte Carlo engine.

When σ_r is varied from 0 to 0.1 (with ρ fixed at 0.25), the estimated price shows a mild upward trend. The confidence intervals remain narrow due to the large number of simulations. This behavior is consistent with the fact that increasing rate volatility affects the distribution of the integrated short rate and its interaction with equity dynamics.

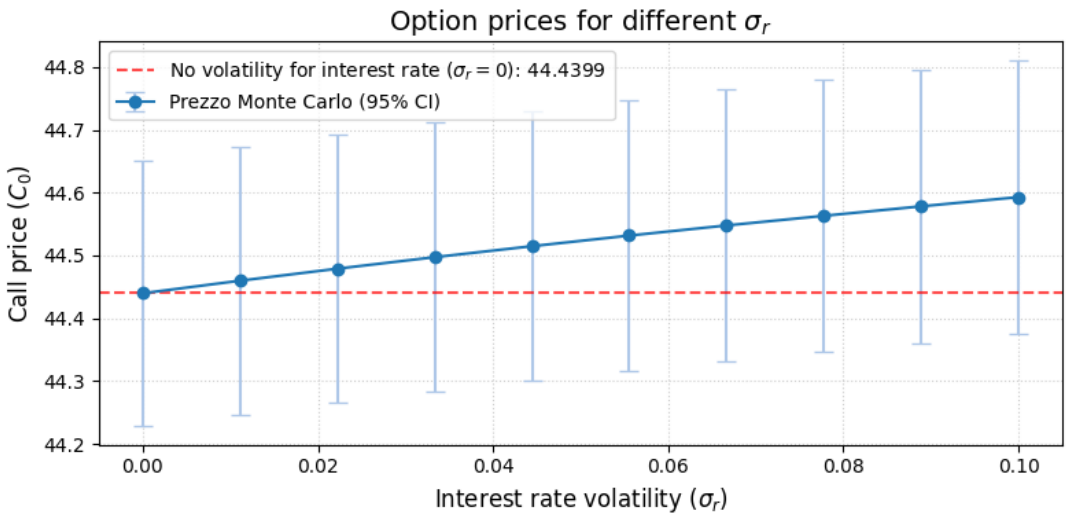

Figure 2: Monte Carlo call prices for different interest-rate volatilities σ_r, with 95% confidence intervals. The dashed line highlights the $\sigma_r = 0$ limit case.

7.1 Interpretation of the interest-rate volatility effect across correlation levels

Changing the short-rate volatility σ_r affects the option price because it increases the randomness of the integrated short rate (and thus the stochastic discount factor) and, through correlation, it also influences the dispersion of the equity component driven by r_t. Importantly, the impact of σ_r depends on the correlation level: the closed-form variance (7) contains both a term proportional to $\rho\,\sigma_r$ and a term proportional to σ_r^2. As a consequence, for $\rho > 0$ the covariance channel tends to amplify the effective variance and increase the call value, while for $\rho < 0$ it tends to reduce it.

This dependence explains why the price increases with σ_r in Figure 2 under the baseline positive correlation, and why the sign of the slope can change when ρ is negative, as illustrated by the joint analysis in Figure 4.

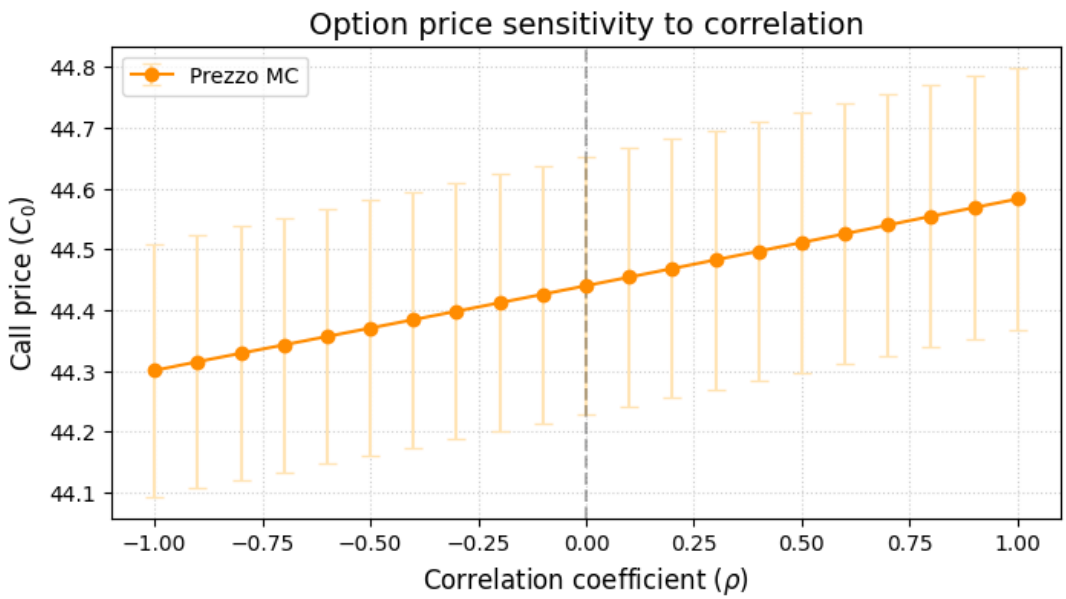

Figure 3: Sensitivity of the call price to the correlation coefficient ρ between equity and rate shocks. Error bars represent 95% confidence intervals.

To make this interaction explicit, we also examine the joint effect of σ_r and ρ by plotting call prices against σ_r for representative correlations $\rho = -0.75, 0, 0.75$. The results show that the sign of ρ can reverse the impact of rate volatility: with positive correlation, larger σ_r increases the call value, while with negative correlation it decreases it. The nearly flat behavior around $\rho = 0$ provides a natural reference case where covariance effects are absent.

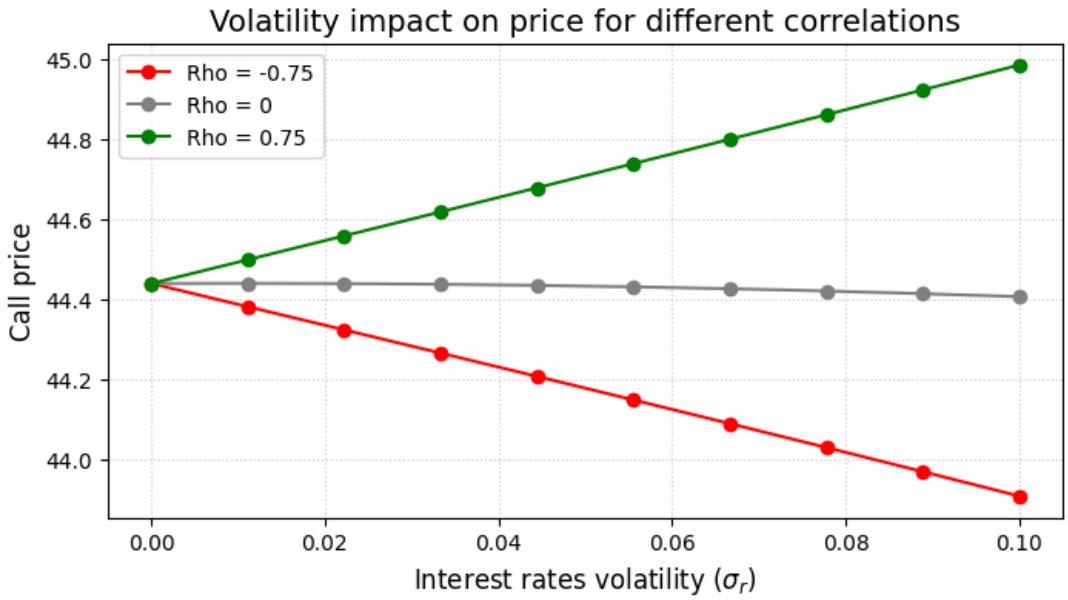

Figure 4: Joint impact of interest-rate volatility σ_r on the call price for different correlation levels ρ. Positive correlation amplifies the price as σ_r increases, while negative correlation reduces it.

7.2 Interpretation of the correlation effect

The monotonic increase of the call premium with respect to ρ can be understood from the effective variance Σ^2 appearing in the closed-form benchmark (7). In particular, ρ enters (7) through the term

$$\frac{2\sigma_r \sigma_S \rho}{\kappa^2}\left(\kappa T - 1 + e^{-\kappa T}\right),$$

and since $\kappa T - 1 + e^{-\kappa T} > 0$ for $T > 0$ and $\kappa > 0$, this contribution is increasing in ρ. Therefore, larger correlation increases the effective dispersion of the (discounted) terminal equity payoff.

From a financial perspective, a European call is vega-positive, meaning its value increases with (effective) volatility. Hence, when ρ increases and Σ^2 increases, the option price increases as well, consistently with the Monte Carlo evidence reported in Figure 3.

8 Empirical illustration with market data

To illustrate applicability to real inputs, the model is also used to price a call on a traded stock by extracting S_0 and an annualized historical volatility estimate for σ_S from one year of daily prices (Yahoo Finance). The interest-rate parameters are set to plausible values and the pricing is computed via both Monte Carlo and the closed-form benchmark.

For Apple (AAPL) with a strike chosen close to the current spot (approximately at-the-money), the inputs lead to a Monte Carlo price around 37.66 with a 95% confidence interval of roughly $[37.26, 38.05]$, while the closed-form value is about 37.93. The absolute difference remains within

the Monte Carlo statistical band. This experiment confirms that, even with empirically estimated equity volatility and a simplified choice of rate parameters, the numerical estimator remains consistent with the analytical formula. In practical calibration, the correlation ρ and the rate parameters would be estimated from joint equity and yield (or short-rate proxy) data, but the main objective here is to demonstrate the end-to-end implementation and validation of the hybrid pricing framework.

9 Conclusion

A Monte Carlo pricing algorithm was implemented for a European call option in a hybrid model where the equity drift is driven by a mean-reverting stochastic short rate. The method combines an Euler discretization for the Vasicek short rate with a conditionally exact lognormal step for the equity, and applies pathwise stochastic discounting via a numerical approximation of $\int_0^T r_t \, dt$. The resulting Monte Carlo estimate matches the closed-form benchmark within a tight confidence interval, supporting correctness and numerical stability. Sensitivity analysis shows that both the interest-rate volatility and the correlation between equity and rate shocks can materially affect the option premium, and that their interaction can change the direction of the volatility effect. An empirical illustration using market data further demonstrates the practical usability of the approach.